

MEAN STACK

DURATION- 4 MONTH

HTML

MODULE 1: WEB PROGRAMMING INTRODUCTION

- Architecture of a website
- Different technologies in making the website
- Web Development Introduction

MODULE 2: HTML-INTRODUCTION

- History of HTML
- What you need to do to get going and make your first HTML page
- What are HTML Tags and Atributes?
- HTML Tag vs. Element
- HTML Atributes

MODULE 3: HTML-HEADERS

- Title
- Base
- Link
- Styles
- Script
- Meta

Module: HTML-Basic formatting tags

- HTML Basic Tags
- HTML Formang Tags
- HTML Color Coding

MODULE 5: HTML-GROUPING USING DIV SPAN

- Div Tag
- Span Tags

MODULE 6: HTML SEMANTIC ELEMENTS

- <arcle>
- <aside>
- <details>
- <figure>
- header>, <footer>
- <main>
- <mark>
- <nav>
- <section>
- <summary>
- <time>

MODULE 7: HTML-LISTS

- Unordered Lists
- Ordered Lists
- Defination list

MODULE 8: HTML-IMAGES

- Image and Image Mapping
- Background Images

MODULE 9: HTML-HYPERLINK

- Anchor Tag
- URL - Uniform Resource Locator
- Relative and Absolute Address

MODULE 10: HTML-TABLE

- < table >
- < th >
- < tr >
- < td >
- < caption >
- < the ad >
- < t body >
- < toot >
- < cogroup >
- < col >

MODULE 11: HTML-I FRAME

- Using I frame as the Target

MODULE 12: HTML-FORM

- HTML Forms
- HTML Form Aributes
- HTML Form Elements
- HTML Input Types
- HTML Input Aributes
- HTML Input form Aributes
- < text area >
- < button >
- < select >
- < label>

MODULE 13: HTML GRAPHICS

- HTML Canvas Graphics
- HTML SVG Graphics

MODULE 14: HTML MEDIA

- HTML Media
- HTML Video & Audio
- HTML Plug-in
- HTML You Tube

MODULE 15: HTML APIS

- HTML Geolocation
- HTML Drag/Drop
- HTML Web Storage & Workers
- HTML SSE

CSS

MODULE 1: CSS INTRODUCTION

- What is CSS?
- Why Use CSS?
- CSS Syntax

MODULE 2: CSS CORE PROPERTIES

- CSS Color
- CSS Backgrounds
- CSS Box Model
- CSS Borders
- CSS Margins
- CSS Paddings
- CSS Box Sizing

MODULE 3: CSS UNITS

- Absolute Units, Relative Units
- Max-Width, Min-Width

MODULE 4: CSS STYLING

- CSS Text
- CSS Fonts
- CSS Outline
- CSS Alignment
- Important

MODULE 5: STYLING ELEMENTS

- Links
- Lists
- Dropdowns
- Tables
- Images
- Image-Sprite
- Image-Filters
- Clip-Path
- Forms

MODULE 6: NAVBAR

- Vercal-Navbar
- Horizontal-Navbar

MODULE 7: CSS SELECTORS

- Simple Selectors
- Combinator selectors
- Pseudo-class selectors
- Pseudo-elements selectors
- Aribute selectors

MODULE 8: CSS POSITIONS

- Position
- Z-Index
- Float
- Clear
- Overflow

MODULE 9: CSS LAYOUTS

- Display
- Block
- Inline, Inline-Block
- Grid
- Flex-Box

MODULE 10: CSS TRANSFORMATION

- CSS 2D & 3D Transforms
- CSS Transition, CSS Animations

MODULE 11: CSS RESPONSIVE

- Responsive Introduction
- Responsive Viewport & Breakpoint
- Media Queries

JAVASCRIPT

MODULE 1: INTRODUCTION

- What is JavaScript?
- Why Use JavaScript?
- Syntax
- Statements
- Comments

MODULE 2: LANGUAGE SYNTAX

- Data Types
- Variable Declarations
- Var vs. Let
- Constant
- Dynamic Type

- Type of
- Type Conversion
- Objects
- Arrays

MODULE 3: OPERATORS

- JavaScript Operators.
- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- Equality Operators
- Ternary Operators
- Logical Operators
- Bitwise Operators

MODULE 4: CONTROL STATEMENT

- if
- else
- if/else
- switch/case

MODULE 5: LOOP STATEMENT

- for
- for/in
- for/off
- while
- do/while
- Infinite loop
- Break
- continue

MODULE 6: JAVASCRIPT HTML DOM

- HTML Document object Model
- DOM Methods
- DOM Documents
- DOM Elements
- DOM HTML
- DOM Forms
- DOM CSS
- DOM Events
- DOM Event Listener
- DOM Navigation
- DOM Nodes
- DOM Collections
- DOM Node Lists

MODULE 7: JAVASCRIPT HTML BOM

- TML Browser object Model
- JS Window
- JS Screen
- JS Location
- JS History
- JS Navigator
- JS Popup Alert
- JS Timing
- JS Cookies

MODULE 8: ARRAY

- Array Introduction
- Adding, Deleng Modifying, Searching & Emptying Element
- Combining and Slicing Array
- Spread Operator
- Looping with Array
- Filtering with Array
- Mapping with Array
- Reducing an Array

MODULE 9: FUNCTIONS

- Function Declarations.
- Hoisng
- Arguments
- The Rest Operators
- Default Parameters
- Geer and Seers
- Try and Catch
- Local vs. Global Scope
- this keyword

MODULE 10: OBJECTS

- Basics
- Factory Functions
- Constructor
- Dynamic Nature of Object
- Functions are Objects
- Value vs. Reference Type
- Enumerang Properes of an Object
- Cloning an Object
- String
- Date

MODULE 11: BUILT-IN OBJECTS

- Number
- Math
- String
- Array
- Date

- Boolean
- Regex

MODULE 12: EVENTS

- Introduction
- Mouse & Keyboard Events
- Form Events
- Document/Window Event

MODULE 13: INTRODUCTION TO OOP CONCEPT

- Prototype, Module pattern
- Augmenting type
- Closures
- ES6 Introduction
- Let & Const
- Arrow Functions
- Class and Inheritance
- Rest and Map Operators
- Export and Import
- Modules

MODULE 14: JAVASCRIPT WEB API

- API Introduction
- Forms API
- History API
- Storage API
- Worker API
- Fetch API
- Geolocation API

MODULE 15: JAVASCRIPT AJAX

- AJAX Introduction
- AJAX XMLHttpRequest
- AJAX Request
- AJAX Response

MODULE 16: JAVASCRIPT JSON

- JSON Introduction
- JSON Syntax
- JSON Data Types
- JSON Parse
- JSON Stringify
- JSON Objects
- JSON Arrays

JQUERY

- Getting started with jQuery
- Selecting elements
- Manipulating the page
- Traversing the DOM and chaining
- JQuery Utility Methods
- Handling events and event delegation
- AJAX, JSON and Deferred
- Enhancing with Animation effects
- Grids, Tables with AJAX, Pagination,
- jQuery UI
- jQuery Best Practices

BOOTSTRAP 5

MODULE 1: INTRODUCTION

- Introduction
- Quick Styles Aer Dark
- Our First Web Page
- What's inside?
- Scripts & Styles
- Core concepts
- Summary

MODULE 2: LAYOUT WITH BOOTSTRAP

- Introduction
- Grid Layouts
- Simple Layout
- Fixed Grids
- Fluid Grids
- Responsive Design
- Responsive Ulies
- Inspiration
- Summary

MODULE 3: EVERYDAY BOOTSTRAP

- Introduction
- Typography
- Tables
- Forms
- Buttons
- Images & Icons
- Summary

MODULE 4: BOOTSTRAP COMPONENT

- Introductions
- Drop Down Menu
- Buttons with Menus
- Tabs & Pil
- The Navbars
- Heros, Badgets, Labels and media
- Summary

MODULE 5: BOOTSTRAP AND JAVASCRIPT

- Introduction
- More Buttons
- Modals
- Toolps And Popovers
- Carousel
- Summary

GITHUB

- Real-Time environmental setup with GitHub

ANGULAR

MODULE-1 INTRODUCTION TO ANGULAR

- What is Angular?
- Advantages of Angular
- Disadvantages of Angular
- Key Features of Angular
- Comparison between Angular & other frameworks

MODULE-2 INTRODUCTION TO TYPESCRIPT

- Why Use Typescript
- Basic Types
- Classes & Interfaces
- Type Definitions
- Compiling Typescript

MODULE-3 ENVIRONMENT SETUP

- Node/ NPM
- Angular CLI
- Application File Structure
- Code Editors

MODULE - 4 MODULES

- Why use Modules
- Ng Module
- Declarations
- Providers
- Imports
- Bootstrapping
- The Core Module
- Shared Modules

MODULE-5 COMPONENTS

- Introduction to Components
- Components Architecture Pattern
- Decorator Metadata
- State & Behavior

MODULE-6 TEMPLATES

- Inline vs External
- Template Expressions
- Data Bindings
- *ngIf else, *ngFor and *ngSwitch
- Built-in Structural Directives
- Built-in Attribute Directives

MODULE-7 CUSTOM DIRECTIVES

- Create your own Structural Directives
- Create your own Attribute Directives

MODULE-8 PIPES

- Built-in Pipes
- Custom Pipes

MODULE-9 SERVICES

- Introduction to services
- Building a Service

MODULE-10 DEPENDENCY INJECTION

- Introduction to dependency Injection
- Injectors & Providers
- Registering Providers

MODULE- 11 LIFECYCLE HOOKS

- Component Lifecycle
- All Lifecycle hooks

MODULE-12 ROUTING

- The component router
- Defining Router
- Navigation
- Route Params
- Child Routes

MODULE-13 ADVANCED ROUTING

- Route Guard

MODULE-14 TEMPLATE-DRIVEN FORMS

- Introduction to forms
- Template-driven forms
- Validation

MODULE-15 MODEL-DRIVEN FORMS

- Introduction to Reactive Forms
- Form group & form control
- Validators
- Custom Validation

MODULE-16 ASYNCHRONOUS OPERATIONS

- Introduction to Async
- Observables
- Observables vs. Promise
- HTTP Request / Response

MODULE-17 COMMUNICATION BETWEEN COMPONENTS

- @Input()
- @Output()
- @View Child()
- Shared Service

MODULE-18 SERVER-SIDE RENDERING

- Introduction to server-side Rendering
- Angular Universal

MODULE-19 ANGULAR TESTING

- Introduction to testing
- Unit Testing
- Lazy Loading
- Angular Material
- Prime NG Implementation

MODULE-20 THIRD-PARTY NPM PACKAGE

- Social Login Authentication
- Cookies, Local Storage, Session Storage
- Angular File Upload
- Angular Pagination
- JWT Token
- IMPLEMENTATION OF MINI APPLICATION
- DEPLOYMENT OF ANGULAR APPLICATION

NODE-JS AND EXPRESS FRAMEWORK

MODULE-1 INTRODUCTION TO NODE.JS

- Introduction to Node.js
- History of Node.js
- Introduction to io.js
- What is Node.js Foundation
- V8 Java Script Engine
- Why Server-Side Java Script?

MODULE-2 GETTING STARTED WITH NODE.JS

- Node.js Architecture
- JavaScript Event Loop
- Node.js vs. others Server-Side Frameworks
- Node.js Application Area
- Advantages of Node.js
- Limitations of Node.js

MODULE-3 JAVASCRIPT WITH NODE.JS

- Writing Asynchronous Code
- Blocking vs. Non-Blocking Code

MODULE-4 MODULES

- Understanding Modules
- Built-In Modules
- Creating Module
- Exporting Module
- Importing Modules Node.js CLI and NPM
- Understanding CLI
- Node's Package Manager: NPM
- Local Packages and Global Packages
- Installing, Updating and Removing Module using NPM
- Understanding package.json file

MODULE-5 Node.JS PACKAGE AND PUBLISHING

- Creating a Node Package
- Publishing Package
- Using published package

MODULE-6 PACKAGE

- Introduction of NPM Package
- Express NPM Package
- Multer NPM Package
- Node Mail NPM Package
- Boot Stripe NPM Package integration
- JSONWEBTOKEN NPM Integration
- Introduction of crypto-JS

MODULE-7 CODE DEBUGGING

- Built-In Debugger
- Node Inspector
- IDE Debuggers

MODULE-8 EXCEPTIONS HANDLING

- Try. Catch
- Call back
- Event Emitters

MODULE-9 EVENTS

- Understanding Events
- Event Emitter class
- Emitting Event
- Listening Event

MODULE-10 STREAMS

- Creating streams
- Streams types
- Readable Stream
- Writable Stream
- Stream Pipe
- Creating Buffer
- Buffer decoding

MODULE-11 FILE SYSTEM AND PATH MODULE

- Introduction to fs Module
- The fs modules operations
- Manipulating Files Create, Open, Read, Write, Close, Rename, Delete
- Manipulating Directives Create, Read, Remove
- The path Module
- Building a Web server
- HTTP Request Methods, Headers
- Response Codes and Headers

MODULE-12 EXPRESS FRAMEWORK

- Understanding Express
- Installing Express
- Creating and Running Express App

MODULE-13 ROUTING AND VIEW ENGINES

- Router Object
- Router Object Methods
- View Engine -Handlebars
- Response Methods

MODULE-14 MIDDLEWARE AND REQUEST PROCESSING

- Middleware
- Express Middleware
- Express Request Processing

MONGODB

MODULE -1

- Understanding NoSQL DB
- NoSQL vs. SQL DB
- Understanding Mongo DB
- Introduction of mongoose Atlas
- MongoDB Data types
- MongoDB Shell Commands
- Understanding db, collection & document
- Understanding Embedded documents
- Querying database Tools & API
- MongoDB Tools
- Introduction to Mongo Chef
- Mongo Chef for database operations

MODULE-2 INDEXING AND RELATIONSHIPS

- Types of Indexes
- Creating an Indexes
- Dropping an Indexes
- Defining Relationships between Documents

MODULE-3 MONGO DB ODM -MONGOOSE AND POSTMAN

- Introduction to Mongoose
- Exploring fundamentals of Mongoose
- Mongoose Models
- Mongoose Data Types
- Mongoose Relationships
- Mongoose CRUD operations

MODULE-4 REST API WITH MONGOOSE, MONGODB AND POSTMAN

- Creating REST API using Express and Mongoose
- IMPLEMENTING THE PROJECT DEPLOYMENT OF MEAN APPLICATION**